Question ID: deff8a2f

Question

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

Answer

A. 60π

B. 120π

C. 600π

D. $2{,}400\pi$

Question ID: 502d9690

Question

Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is 648 square inches, and the area of rectangle $EFGH$ is 72 square inches. The length of the longest side of rectangle $ABCD$ is 36 inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

Answer

A. 4

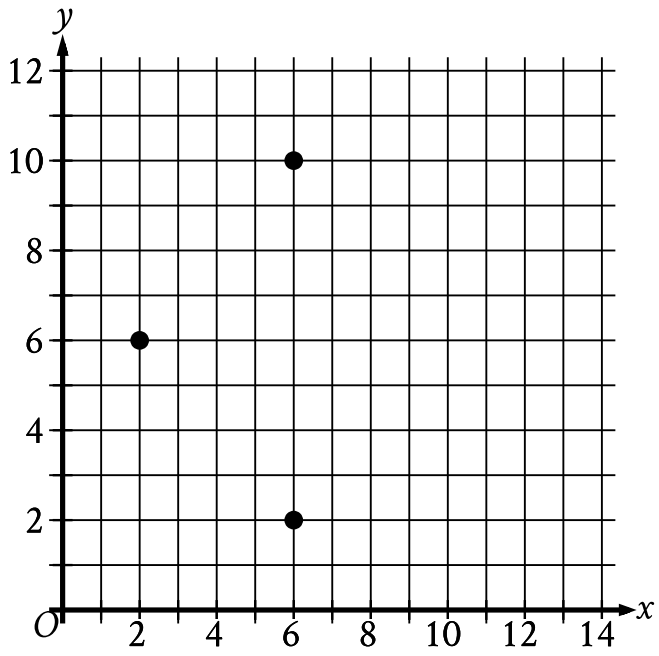
B. 9

C. 12

D. 36

Question ID: b2528e6b

Question



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k?

Question ID: 5b2b8866

Question

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 20%. What is the area of the copy, in square inches?

Question ID: 6708546e

Question

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is 5 square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

Answer

A. 7

B. 10

C. 20

D. 25

Question ID: b0dc920d

Question

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V, in cubic inches, of such containers, where r is the radius, in inches?

Answer

A. $V = 4\pi r^3$

B. $V = \pi(2r)^3$

C. $V = \pi r^2 + 4\pi r$

D. $V = \pi r^3 + 4\pi r^2$

Question ID: 310c87fe

Question

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

Answer

A. 18

B. 27

C. 36

D. 81

Question ID: 459dd6c5

Question

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF. The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF?

Question ID: c984f1a5

Question

A hemisphere is half of a sphere. If a hemisphere has a radius of 27 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

Answer

A. $1{,}500$

B. $6{,}100$

C. $30{,}900$

D. $41{,}200$

Question ID: f7e626b2

Question

The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

Answer

A. 30

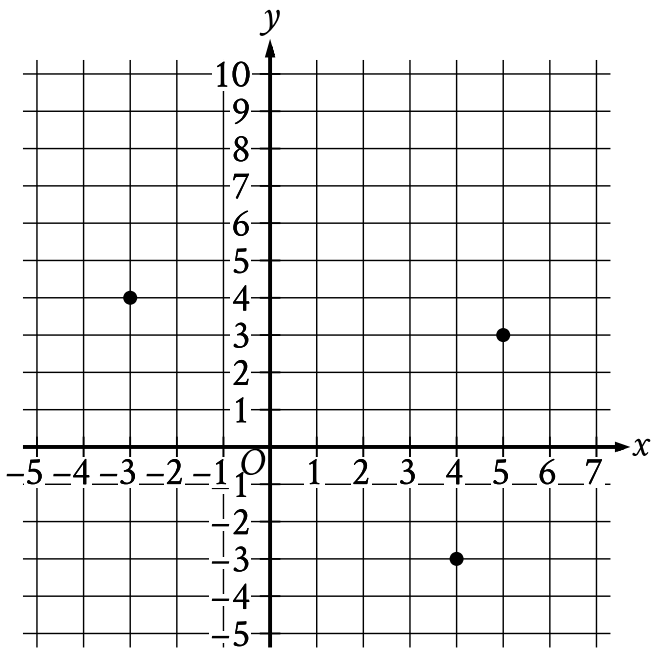
B. 74

C. 120

D. 148

Question ID: eb70d2d0

Question



What is the area, in square units, of the triangle formed by connecting the three points shown?

Question ID: 93de3f84

Question

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

Answer

A. 11

B. 22

C. 44

D. 66

Question ID: dc71597b

Question

A right circular cone has a volume of $\frac{1}{3}\pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

Answer

A. $\frac{1}{3}$

B. $\frac{1}{\sqrt{3}}$

C. $\sqrt{3}$

D. 3